

・三角関数の不定積分 ～ $\tan \frac{x}{2} = t$ と置換～

(例) 次の不定積分を求めよ。

$$\int \frac{1}{3 \sin x + 4 \cos x} dx$$

point

どうしようもないとき

$\tan \frac{x}{2} = t$ において、有理関数に帰着する

$\tan \frac{x}{2} = t$ とおくと

$$\frac{1}{2} \frac{1}{\cos^2 \frac{x}{2}} dx = dt$$

より

$$\frac{1}{2} (1 + \tan^2 \frac{x}{2}) dx = dt \quad \text{つまり} \quad dx = \frac{2}{1+t^2} dt$$

また

$$\begin{aligned} \cos x &= 2 \cos^2 \frac{x}{2} - 1 \\ &= \frac{2}{1 + \tan^2 \frac{x}{2}} - 1 \\ &= \frac{1-t^2}{1+t^2} \end{aligned}$$

$$\begin{aligned} \sin x &= 2 \sin \frac{x}{2} \cos \frac{x}{2} \\ &= 2 \tan \frac{x}{2} \cos^2 \frac{x}{2} \\ &= 2 \tan \frac{x}{2} \frac{1}{1 + \tan^2 \frac{x}{2}} \\ &= \frac{2t}{1+t^2} \end{aligned}$$

であるから

$$\begin{aligned} (\text{与式}) &= \int \frac{1}{3 \frac{2t}{1+t^2} + 4 \frac{1-t^2}{1+t^2}} \cdot \frac{2}{1+t^2} dt \\ &= - \int \frac{1}{2t^2 - 3t - 2} dt \\ &= - \int \frac{1}{(2t+1)(t-2)} dt \\ &= \frac{1}{5} \int \left(\frac{2}{2t+1} - \frac{1}{t-2} \right) dt \\ &= \frac{1}{5} (\log |2t+1| - \log |t-2|) + C \\ &= \frac{1}{5} \log \left| \frac{2t+1}{t-2} \right| + C \\ &= \frac{1}{5} \log \left| \frac{2 \tan \frac{x}{2} + 1}{\tan \frac{x}{2} - 2} \right| + C \quad (C \text{ は積分定数}) \end{aligned}$$

$$\left\{ \begin{array}{l} -\frac{1}{(2t+1)(t-2)} = \frac{a}{2t+1} + \frac{b}{t-2} \\ a(t-2) + b(2t+1) = -1 \\ \begin{cases} -\frac{5}{2}a = -1 \\ 5b = -1 \end{cases} \therefore \begin{cases} a = \frac{2}{5} \\ b = -\frac{1}{5} \end{cases} \end{array} \right.$$

(参考) $\sin x$ の求め方

$$\begin{aligned} \tan x &= \frac{2 \tan \frac{x}{2}}{1 - \tan^2 \frac{x}{2}} \\ &= \frac{2t}{1-t^2} \end{aligned}$$

$$\begin{aligned} \cos x &= 2 \cos^2 \frac{x}{2} - 1 \\ &= \frac{2}{1 + \tan^2 \frac{x}{2}} - 1 \\ &= \frac{1-t^2}{1+t^2} \end{aligned}$$

より

$$\begin{aligned} \sin x &= \tan x \cos x \\ &= \frac{2t}{1+t^2} \end{aligned}$$